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INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des sciences et des technologies (FST)

Administration Système_ TD N° 3

Préparé par : Brian ANTOINE

Présenté à: M. Ismaël SAINT AMOUR

Niveau : L3

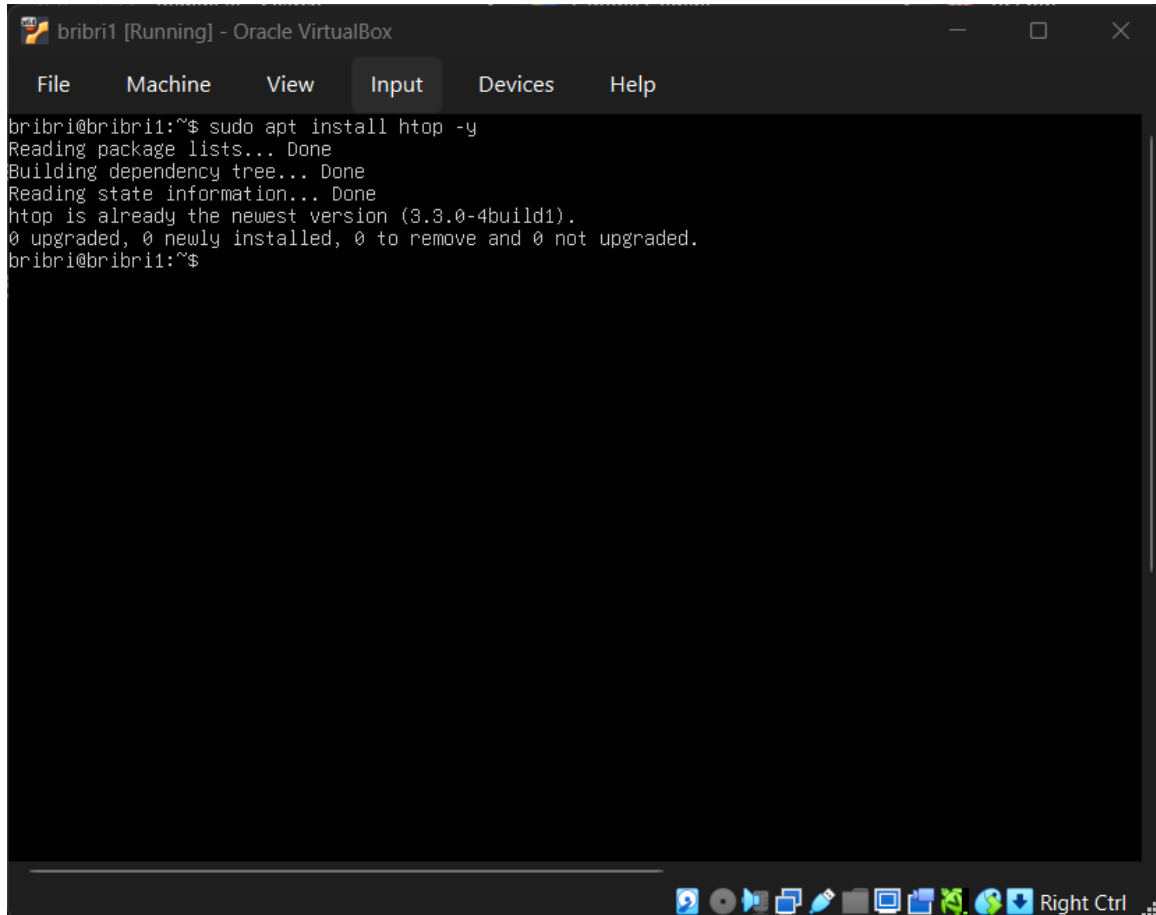
Date: 22/03/2026

L'objectif de ce TD est de :

1. Surveiller les ressources système
2. Analyser les logs avec journalctl
3. Comprendre les logs applicatifs
4. Mettre en place une supervision simple
5. Détecter un incident
6. Mettre en place une supervision simple automatisée
7. Comprendre la rotation des logs.

1. Supervision du système en temps réel

- Surveillance des ressources en temps réel



The image shows a terminal window titled "bribri1 [Running] - Oracle VirtualBox". The terminal output shows the command `sudo apt install htop -y` being executed. The output indicates that the package lists are read, the dependency tree is built, and the state information is read. It then states that htop is already the newest version (3.3.0-4build1) and that 0 packages were upgraded, 0 newly installed, 0 to be removed, and 0 not upgraded. The prompt returns to `bribri@bribri1:~$`. The terminal window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". The bottom of the window shows a taskbar with various icons and the text "Right Ctrl".

```
bribri@bribri1:~$ sudo apt install htop -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
htop is already the newest version (3.3.0-4build1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
bribri@bribri1:~$
```

```

bribr1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

00:00 2.0s Tasks: 22, 22 thr, 88 kthr: 2 running
Mem: 1.23s Load average: 0.00 0.03 0.06
Sub: 19PM/3.82G Uptime: 00:18:45
OK/OK

Main PID USER PRI NI VIRT RES SHR S CPU%MEMt TIME+ Command
2555 bribr1 20 0 8688 4388 3676 S 0.7 0.1 0:30.49 top
1 root 20 0 25432 15780 9580 S 0.0 0.3 0:03.41 /sbin/init splash noprompt noshell automatic-ubiquity
313 root 15 -1 58444 16432 15312 S 0.0 0.4 0:00.66 /usr/lib/systemd/systemd-journald
359 root RT 0 282M 27324 8760 S 0.0 0.7 0:00.44 /sbin/multipathd -d -s
380 root 20 0 282M 27324 8760 S 0.0 0.7 0:00.00 /sbin/multipathd -d -s
382 root RT 0 282M 27324 8760 S 0.0 0.7 0:00.00 /sbin/multipathd -d -s
383 root RT 0 282M 27324 8760 S 0.0 0.7 0:00.00 /sbin/multipathd -d -s
384 root RT 0 282M 27324 8760 S 0.0 0.7 0:00.00 /sbin/multipathd -d -s
385 root RT 0 282M 27324 8760 S 0.0 0.7 0:01.06 /sbin/multipathd -d -s
386 root RT 0 282M 27324 8760 S 0.0 0.7 0:00.00 /sbin/multipathd -d -s
404 root 20 0 29080 7816 5680 S 0.0 0.2 0:00.58 /usr/lib/systemd/systemd-udev
424 systemd-re 20 0 21592 13060 10744 S 0.0 0.3 0:00.36 /usr/lib/systemd/systemd-resolved
889 systemd-ne 20 0 19012 9716 8544 S 0.0 0.2 0:00.12 /usr/lib/systemd/systemd-networkd
851 root 20 0 4236 2680 2420 S 0.0 0.1 0:00.03 /usr/sbin/cron -f -P
852 messagebus 20 0 9980 5760 4656 S 0.0 0.1 0:00.38 dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-o
855 polkitd 20 0 300M 8140 7236 S 0.0 0.2 0:00.12 /usr/lib/polkit-1/polkitd --no-debug
865 root 20 0 18132 8880 7776 S 0.0 0.2 0:00.21 /usr/lib/systemd/systemd-logind
869 root 20 0 457M 13588 11436 S 0.0 0.3 0:00.20 /usr/libexec/udisks2/udisks
879 root 20 0 457M 13588 11436 S 0.0 0.3 0:00.07 /usr/libexec/udisks2/udisks
880 root 20 0 457M 13588 11436 S 0.0 0.3 0:00.00 /usr/libexec/udisks2/udisks
882 root 20 0 457M 13588 11436 S 0.0 0.3 0:00.01 /usr/libexec/udisks2/udisks
901 root 20 0 104M 23036 13580 S 0.0 0.6 0:00.29 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
904 polkitd 20 0 300M 8140 7236 S 0.0 0.2 0:00.07 /usr/lib/polkit-1/polkitd --no-debug
905 polkitd 20 0 300M 8140 7236 S 0.0 0.2 0:00.00 /usr/lib/polkit-1/polkitd --no-debug
921 root 20 0 382M 13120 11040 S 0.0 0.3 0:00.25 /usr/sbin/ModemManager
922 root 20 0 457M 13588 11436 S 0.0 0.3 0:00.00 /usr/libexec/udisks2/udisks
930 root 20 0 457M 13588 11436 S 0.0 0.3 0:00.00 /usr/libexec/udisks2/udisks
943 root 20 0 382M 13120 11040 S 0.0 0.3 0:00.00 /usr/sbin/ModemManager
945 root 20 0 382M 13120 11040 S 0.0 0.3 0:00.00 /usr/sbin/ModemManager
947 root 20 0 5944 4872 4860 S 0.0 0.1 0:00.14 /bin/login -p --
950 root 20 0 382M 13120 11040 S 0.0 0.3 0:00.00 /usr/sbin/ModemManager
964 root 20 0 104M 23036 13580 S 0.0 0.6 0:00.00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
1085 systemd-t 20 0 91028 7984 7080 S 0.0 0.2 0:00.13 /usr/lib/systemd/systemd-timesyncd
1087 systemd-t 20 0 91028 7984 7080 S 0.0 0.2 0:00.00 /usr/lib/systemd/systemd-timesyncd
1150 bribr1 20 0 29028 11432 9352 S 0.0 0.3 0:00.45 /usr/lib/systemd/systemd-user
1151 bribr1 20 0 21152 3568 1844 S 0.0 0.1 0:00.00 (sd-pam)
1202 bribr1 20 0 8656 5636 3908 S 0.0 0.1 0:00.15 -bash
1247 bribr1 20 0 11944 6008 3776 S 0.0 0.1 0:00.23 top
2132 syslogd 20 0 217M 5400 4556 S 0.0 0.1 0:00.05 /usr/sbin/rsyslogd -n -iNONE
2134 syslogd 20 0 217M 5400 4556 S 0.0 0.1 0:00.00 /usr/sbin/rsyslogd -n -iNONE

Main PID USER PRI NI VIRT RES SHR S CPU%MEMt TIME+ Command
(3) Stopped top
bribr1:bribr1:~$ df -h
Filesystem Size Used Avail Use% Mounted on
tmpfs 392M 1.1M 391M 1% /run
/dev/sda2 25G 2.7G 21G 12% /
tmpfs 2.0G 0 2.0G 0% /dev/shm
tmpfs 5.0M 0 5.0M 0% /run/lock
tmpfs 392M 12K 392M 1% /run/user/1000
bribr1:bribr1:~$

```

➤ Vérification de la mémoire : df -h

```

bribr1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

1 root 20 0 25432 15780 9580 S 0.0 0.3 0:03.41 /sbin/init splash noprompt noshell automatic-ubiquity
15 -1 58444 16432 15312 S 0.0 0.4 0:00.66 /usr/lib/systemd/systemd-journald
359 root RT 0 282M 27324 8760 S 0.0 0.7 0:00.45 /sbin/multipathd -d -s
380 root 20 0 282M 27324 8760 S 0.0 0.7 0:00.00 /sbin/multipathd -d -s
382 root RT 0 282M 27324 8760 S 0.0 0.7 0:00.00 /sbin/multipathd -d -s
383 root RT 0 282M 27324 8760 S 0.0 0.7 0:00.00 /sbin/multipathd -d -s
384 root RT 0 282M 27324 8760 S 0.0 0.7 0:00.00 /sbin/multipathd -d -s
385 root RT 0 282M 27324 8760 S 0.0 0.7 0:01.06 /sbin/multipathd -d -s
386 root RT 0 282M 27324 8760 S 0.0 0.7 0:00.00 /sbin/multipathd -d -s
404 root 20 0 29080 7816 5680 S 0.0 0.2 0:00.58 /usr/lib/systemd/systemd-udev
424 systemd-re 20 0 21592 13060 10744 S 0.0 0.3 0:00.36 /usr/lib/systemd/systemd-resolved
889 systemd-ne 20 0 19012 9716 8544 S 0.0 0.2 0:00.12 /usr/lib/systemd/systemd-networkd
851 root 20 0 4236 2680 2420 S 0.0 0.1 0:00.03 /usr/sbin/cron -f -P
852 messagebus 20 0 9980 5760 4656 S 0.0 0.1 0:00.38 dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-o
855 polkitd 20 0 300M 8140 7236 S 0.0 0.2 0:00.12 /usr/lib/polkit-1/polkitd --no-debug
865 root 20 0 18132 8880 7776 S 0.0 0.2 0:00.21 /usr/lib/systemd/systemd-logind
869 root 20 0 457M 13588 11436 S 0.0 0.3 0:00.07 /usr/libexec/udisks2/udisks
879 root 20 0 457M 13588 11436 S 0.0 0.3 0:00.00 /usr/libexec/udisks2/udisks
882 root 20 0 457M 13588 11436 S 0.0 0.3 0:00.01 /usr/libexec/udisks2/udisks
901 root 20 0 104M 23036 13580 S 0.0 0.6 0:00.29 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
904 polkitd 20 0 300M 8140 7236 S 0.0 0.2 0:00.07 /usr/lib/polkit-1/polkitd --no-debug
905 polkitd 20 0 300M 8140 7236 S 0.0 0.2 0:00.00 /usr/lib/polkit-1/polkitd --no-debug
921 root 20 0 382M 13120 11040 S 0.0 0.3 0:00.25 /usr/sbin/ModemManager
922 root 20 0 457M 13588 11436 S 0.0 0.3 0:00.00 /usr/libexec/udisks2/udisks
930 root 20 0 457M 13588 11436 S 0.0 0.3 0:00.00 /usr/libexec/udisks2/udisks
943 root 20 0 382M 13120 11040 S 0.0 0.3 0:00.00 /usr/sbin/ModemManager
945 root 20 0 382M 13120 11040 S 0.0 0.3 0:00.00 /usr/sbin/ModemManager
947 root 20 0 5944 4872 4860 S 0.0 0.1 0:00.14 /bin/login -p --
950 root 20 0 382M 13120 11040 S 0.0 0.3 0:00.00 /usr/sbin/ModemManager
964 root 20 0 104M 23036 13580 S 0.0 0.6 0:00.00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
1085 systemd-t 20 0 91028 7984 7080 S 0.0 0.2 0:00.13 /usr/lib/systemd/systemd-timesyncd
1087 systemd-t 20 0 91028 7984 7080 S 0.0 0.2 0:00.00 /usr/lib/systemd/systemd-timesyncd
1150 bribr1 20 0 29028 11432 9352 S 0.0 0.3 0:00.45 /usr/lib/systemd/systemd-user
1151 bribr1 20 0 21152 3568 1844 S 0.0 0.1 0:00.00 (sd-pam)
1202 bribr1 20 0 8656 5636 3908 S 0.0 0.1 0:00.15 -bash
1247 bribr1 20 0 11944 6008 3776 S 0.0 0.1 0:00.23 top
2132 syslogd 20 0 217M 5400 4556 S 0.0 0.1 0:00.05 /usr/sbin/rsyslogd -n -iNONE
2134 syslogd 20 0 217M 5400 4556 S 0.0 0.1 0:00.00 /usr/sbin/rsyslogd -n -iNONE

(3) Stopped top
bribr1:bribr1:~$ df -h
Filesystem Size Used Avail Use% Mounted on
tmpfs 392M 1.1M 391M 1% /run
/dev/sda2 25G 2.7G 21G 12% /
tmpfs 2.0G 0 2.0G 0% /dev/shm
tmpfs 5.0M 0 5.0M 0% /run/lock
tmpfs 392M 12K 392M 1% /run/user/1000
bribr1:bribr1:~$

```

2. Analyse des logs :

➤ Logs systemd : journalctl

```
brbri1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Mar 22 23:59:49 brbri1 sudo[1283]: brbri1 : TTYvtty1 : PWG/home/brbri1 : USER=root : COMMAND=/usr/bin/apt update
Mar 22 23:59:49 brbri1 sudo[1283]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 23 00:00:02 brbri1 sudo[1283]: pam_unix(sudo:session): session closed for user root
Mar 23 00:03:16 brbri1 sudo[1190]: launchpadlib-cache-clean.service - Clean up old files in the Launchpadlib cache was skipped because of an unset condition
Mar 23 00:03:47 brbri1 sudo[1620]: brbri1 : TTYvtty1 : PWG/home/brbri1 : USER=root : COMMAND=/usr/bin/apt upgrade
Mar 23 00:03:47 brbri1 sudo[1620]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 23 00:04:29 brbri1 sudo[1620]: pam_unix(sudo:session): session closed for user root
Mar 23 00:05:08 brbri1 sudo[2195]: brbri1 : TTYvtty1 : PWG/home/brbri1 : USER=root : COMMAND=/usr/bin/apt install http -y
Mar 23 00:05:08 brbri1 sudo[2195]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 23 00:05:09 brbri1 sudo[2195]: pam_unix(sudo:session): session closed for user root

[7]+ Stopped Journalctl
brbri1@brbri1:~$ journalctl
Hint: You are currently not seeing messages from other users and the system.
Users in groups 'adm', 'systemd-journal' can see all messages.

Pass -q to turn off this notice.

Mar 22 23:57:56 brbri1 systemd[1190]: Queued start job for default target default.target.
Mar 22 23:57:56 brbri1 systemd[1190]: Created slice app.slice - User Application Slice.
Mar 22 23:57:56 brbri1 systemd[1190]: Started launchpadlib-cache-clean.timer - Clean up old files in the Launchpadlib cache.
Mar 22 23:57:56 brbri1 systemd[1190]: Reached target paths.target - Paths.
Mar 22 23:57:56 brbri1 systemd[1190]: Reached target timers.target - Timers.
Mar 22 23:57:56 brbri1 systemd[1190]: Starting dbus.socket - D-Bus User Message Bus Socket...
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on dirvgnr.socket - GnuPG network certificate management daemon.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on gpg-agent-browser.socket - GnuPG cryptographic agent and passphrase cache (access for web browsers).
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on gpg-agent-extra.socket - GnuPG cryptographic agent and passphrase cache (restricted).
Mar 22 23:57:56 brbri1 systemd[1190]: Starting gpg-agent-ssh.socket - GnuPG cryptographic agent (ssh-agent emulation)...
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on gpg-agent.socket - GnuPG cryptographic agent and passphrase cache.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on keyboard.socket - GnuPG public key management service.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on pk-debconf-helper.socket - debconf communication socket.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on snapd.session-agent.socket - REST API socket for snapd user session agent.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Mar 22 23:57:57 brbri1 systemd[1190]: Listening on gpg-agent-ssh.socket - GnuPG cryptographic agent (ssh-agent emulation).
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target sockets.target - Sockets.
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target basic.target - Basic System.
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target default.target - Main User Target.
Mar 22 23:57:57 brbri1 systemd[1190]: Startup finished in 983ms.
Mar 22 23:59:16 brbri1 sudo[1251]: brbri1 : TTYvtty1 : PWG/home/brbri1 : USER=root : COMMAND=/usr/bin/apt install http -y
Mar 22 23:59:16 brbri1 sudo[1251]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 22 23:59:17 brbri1 sudo[1251]: pam_unix(sudo:session): session closed for user root
Mar 22 23:59:49 brbri1 sudo[1283]: brbri1 : TTYvtty1 : PWG/home/brbri1 : USER=root : COMMAND=/usr/bin/apt update
Mar 22 23:59:49 brbri1 sudo[1283]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 23 00:00:02 brbri1 sudo[1283]: pam_unix(sudo:session): session closed for user root
Mar 23 00:03:16 brbri1 sudo[1190]: launchpadlib-cache-clean.service - Clean up old files in the Launchpadlib cache was skipped because of an unset condition
Mar 23 00:03:47 brbri1 sudo[1620]: brbri1 : TTYvtty1 : PWG/home/brbri1 : USER=root : COMMAND=/usr/bin/apt upgrade
Mar 23 00:03:47 brbri1 sudo[1620]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 23 00:04:29 brbri1 sudo[1620]: pam_unix(sudo:session): session closed for user root
Mar 23 00:05:08 brbri1 sudo[2195]: brbri1 : TTYvtty1 : PWG/home/brbri1 : USER=root : COMMAND=/usr/bin/apt install http -y
Mar 23 00:05:08 brbri1 sudo[2195]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 23 00:05:09 brbri1 sudo[2195]: pam_unix(sudo:session): session closed for user root
lines 1-53/53 (END)
```

➤ Logs système principaux :

```
brbri1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

A start job for unit UNIT has finished successfully.
The job identifier is 5.
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target basic.target - Basic System.
Subject: A start job for unit UNIT has finished successfully
Defined-By: systemd
Support: http://www.ubuntu.com/support

A start job for unit UNIT has finished successfully.
The job identifier is 4.
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target default.target - Main User Target.
Subject: A start job for unit UNIT has finished successfully
Defined-By: systemd
Support: http://www.ubuntu.com/support

A start job for unit UNIT has finished successfully.
The job identifier is 3.
Mar 22 23:57:57 brbri1 systemd[1190]: Startup finished in 983ms.
Subject: User manager start-up is now complete
Defined-By: systemd
Support: http://www.ubuntu.com/support

The user manager instance for user 1000 has been started. All services queued
for starting have been started. Note that other services might still be starting
up or be started at any later time.

Startup of the manager took 903630 microseconds.
Mar 22 23:59:16 brbri1 sudo[1251]: brbri1 : TTYvtty1 : PWG/home/brbri1 : USER=root : COMMAND=/usr/bin/apt install http -y
Mar 22 23:59:16 brbri1 sudo[1251]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 22 23:59:17 brbri1 sudo[1251]: pam_unix(sudo:session): session closed for user root
Mar 22 23:59:49 brbri1 sudo[1283]: brbri1 : TTYvtty1 : PWG/home/brbri1 : USER=root : COMMAND=/usr/bin/apt update
Mar 22 23:59:49 brbri1 sudo[1283]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 23 00:00:02 brbri1 sudo[1283]: pam_unix(sudo:session): session closed for user root
Mar 23 00:03:16 brbri1 sudo[1190]: launchpadlib-cache-clean.service - Clean up old files in the Launchpadlib cache was skipped because of an unset condition
Subject: A start job for unit UNIT has finished successfully
Defined-By: systemd
Support: http://www.ubuntu.com/support

A start job for unit UNIT has finished successfully.
The job identifier is 22.
Mar 23 00:03:47 brbri1 sudo[1620]: brbri1 : TTYvtty1 : PWG/home/brbri1 : USER=root : COMMAND=/usr/bin/apt upgrade
Mar 23 00:03:47 brbri1 sudo[1620]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 23 00:04:29 brbri1 sudo[1620]: pam_unix(sudo:session): session closed for user root
Mar 23 00:05:08 brbri1 sudo[2195]: brbri1 : TTYvtty1 : PWG/home/brbri1 : USER=root : COMMAND=/usr/bin/apt install http -y
Mar 23 00:05:08 brbri1 sudo[2195]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 23 00:05:09 brbri1 sudo[2195]: pam_unix(sudo:session): session closed for user root
lines 127-175/175 (END)
```

```
brbri1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

||| The Job Identifier is 5.
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target basic.target - Basic System.
Subject: A start job for unit UNIT has finished successfully
Defined-By: systemd
Support: http://www.ubuntu.com/support

A start job for unit UNIT has finished successfully.

The Job Identifier is 4.
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target default.target - Main User Target.
Subject: A start job for unit UNIT has finished successfully
Defined-By: systemd
Support: http://www.ubuntu.com/support

A start job for unit UNIT has finished successfully.

The Job Identifier is 3.
Mar 22 23:57:57 brbri1 systemd[1190]: Startup finished in 983ms.
Subject: User manager start-up is now complete
Defined-By: systemd
Support: http://www.ubuntu.com/support

The user manager instance for user 1000 has been started. All services needed
for starting have been started. Note that other services might still be starting
up or be started at any later time.

Startup of the manager took 983638 microseconds.
Mar 22 23:59:16 brbri1 sudo[1251]: brbri : TTYvtty : PWG:/home/brbri : USER=root : COMMAND=/usr/bin/apt install http -y
Mar 22 23:59:16 brbri1 sudo[1251]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri(uid=1000)
Mar 22 23:59:17 brbri1 sudo[1251]: pam_unix(sudo:session): session closed for user root
Mar 22 23:59:49 brbri1 sudo[1283]: brbri : TTYvtty : PWG:/home/brbri : USER=root : COMMAND=/usr/bin/apt update
Mar 22 23:59:49 brbri1 sudo[1283]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri(uid=1000)
Mar 22 00:00:02 brbri1 sudo[1283]: pam_unix(sudo:session): session closed for user root
Mar 22 00:03:16 brbri1 systemd[1190]: launchpadlib-cache-clean.service - Clean up old files in the Launchpadlib cache was skipped because of an unmet condition.
Subject: A start job for unit UNIT has finished successfully
Defined-By: systemd
Support: http://www.ubuntu.com/support

A start job for unit UNIT has finished successfully.

The Job Identifier is 22.
Mar 23 00:03:47 brbri1 sudo[1620]: brbri : TTYvtty : PWG:/home/brbri : USER=root : COMMAND=/usr/bin/apt upgrade
Mar 23 00:03:47 brbri1 sudo[1620]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri(uid=1000)
Mar 23 00:04:29 brbri1 sudo[1620]: pam_unix(sudo:session): session closed for user root
Mar 23 00:05:08 brbri1 sudo[2195]: brbri : TTYvtty : PWG:/home/brbri : USER=root : COMMAND=/usr/bin/apt install http -y
Mar 23 00:05:08 brbri1 sudo[2195]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri(uid=1000)
Mar 23 00:05:09 brbri1 sudo[2195]: pam_unix(sudo:session): session closed for user root

(6)+ Stopped journalctl -xe
brbri@brbri1:~$
```

➤ Affichage des logs du dernier démarrage : journalctl -b

```
brbri1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Mar 23 00:00:02 brbri1 sudo[1283]: pam_unix(sudo:session): session closed for user root
Mar 22 00:03:16 brbri1 systemd[1190]: launchpadlib-cache-clean.service - Clean up old files in the Launchpadlib cache was skipped because of an unmet condition.
Mar 23 00:03:47 brbri1 sudo[1620]: brbri : TTYvtty : PWG:/home/brbri : USER=root : COMMAND=/usr/bin/apt upgrade
Mar 23 00:03:47 brbri1 sudo[1620]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri(uid=1000)
Mar 23 00:04:29 brbri1 sudo[1620]: pam_unix(sudo:session): session closed for user root
Mar 23 00:05:08 brbri1 sudo[2195]: brbri : TTYvtty : PWG:/home/brbri : USER=root : COMMAND=/usr/bin/apt install http -y
Mar 23 00:05:08 brbri1 sudo[2195]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri(uid=1000)
Mar 23 00:05:09 brbri1 sudo[2195]: pam_unix(sudo:session): session closed for user root

(9)+ Stopped journalctl
brbri@brbri1:~$ journalctl -b
Hint: You are currently not seeing messages from other users and the system.
      Users in other sessions can see all messages.
      Pass -q to turn off this notice.
Mar 22 23:57:56 brbri1 systemd[1190]: Queued start job for default target default.target.
Mar 22 23:57:56 brbri1 systemd[1190]: Created slice app.slice - User Application Slice.
Mar 22 23:57:56 brbri1 systemd[1190]: Reached target path.target - Paths.
Mar 22 23:57:56 brbri1 systemd[1190]: Reached target timers.target - Timers.
Mar 22 23:57:56 brbri1 systemd[1190]: Starting dbus.socket - D-Bus User Message Bus Socket...
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on dirnrg.socket - GnuPG network certificate management daemon.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on gpg-agent-browser.socket - GnuPG cryptographic agent and passphrase cache (access for web browsers).
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on gpg-agent-extra.socket - GnuPG cryptographic agent and passphrase cache (restricted).
Mar 22 23:57:56 brbri1 systemd[1190]: Starting gpg-agent-ssh.socket - GnuPG cryptographic agent (ssh-agent emulation)...
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on gpg-agent.socket - GnuPG cryptographic agent and passphrase cache.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on keyboard.socket - GnuPG public key management service.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on pk-debconf-helper.socket - debconf communication socket.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on smbd.session-agent.socket - REST API socket for smbd user session agent.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Mar 22 23:57:57 brbri1 systemd[1190]: Listening on gpg-agent-ssh.socket - GnuPG cryptographic agent (ssh-agent emulation).
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target sockets.target - Sockets.
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target basic.target - Basic System.
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target default.target - Main User Target.
Mar 22 23:57:57 brbri1 systemd[1190]: Startup finished in 983ms.
Mar 22 23:59:16 brbri1 sudo[1251]: brbri : TTYvtty : PWG:/home/brbri : USER=root : COMMAND=/usr/bin/apt install http -y
Mar 22 23:59:16 brbri1 sudo[1251]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri(uid=1000)
Mar 22 23:59:17 brbri1 sudo[1251]: pam_unix(sudo:session): session closed for user root
Mar 22 23:59:49 brbri1 sudo[1283]: brbri : TTYvtty : PWG:/home/brbri : USER=root : COMMAND=/usr/bin/apt update
Mar 22 23:59:49 brbri1 sudo[1283]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri(uid=1000)
Mar 22 00:00:02 brbri1 sudo[1283]: pam_unix(sudo:session): session closed for user root
Mar 22 00:03:16 brbri1 systemd[1190]: launchpadlib-cache-clean.service - Clean up old files in the Launchpadlib cache was skipped because of an unmet condition.
Mar 23 00:03:47 brbri1 sudo[1620]: brbri : TTYvtty : PWG:/home/brbri : USER=root : COMMAND=/usr/bin/apt upgrade
Mar 23 00:03:47 brbri1 sudo[1620]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri(uid=1000)
Mar 23 00:04:29 brbri1 sudo[1620]: pam_unix(sudo:session): session closed for user root
Mar 23 00:05:08 brbri1 sudo[2195]: brbri : TTYvtty : PWG:/home/brbri : USER=root : COMMAND=/usr/bin/apt install http -y
Mar 23 00:05:08 brbri1 sudo[2195]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri(uid=1000)
Mar 23 00:05:09 brbri1 sudo[2195]: pam_unix(sudo:session): session closed for user root

(9)+ Stopped journalctl -b
brbri@brbri1:~$
```

➤ Filtrer les erreurs : journalctl -p err

```
brbri1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Mar 22 23:57:56 brbri1 systemd[1190]: Listening on keyboard.socket - GnuPG public key management service.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on pk-debconf-helper.socket - debconf communication socket.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on snapd.session-agent.socket - REST API socket for snapd user session agent.
Mar 22 23:57:56 brbri1 systemd[1190]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Mar 22 23:57:57 brbri1 systemd[1190]: Listening on gpg-agent-ssh.socket - GnuPG cryptographic agent (ssh-agent emulation).
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target sockets.target - Sockets.
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target basic.target - Basic System.
Mar 22 23:57:57 brbri1 systemd[1190]: Reached target default.target - Main User Target.
Mar 22 23:57:57 brbri1 systemd[1190]: Startup finished in 9.02ms.
Mar 22 23:59:16 brbri1 sudo[1251]: brbri : TTY=ttty ; PWD=/home/brbri1 ; USER=root ; COMMAND=/usr/bin/apt install httpd -y
Mar 22 23:59:16 brbri1 sudo[1251]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 22 23:59:17 brbri1 sudo[1251]: pam_unix(sudo:session): session closed for user root
Mar 22 23:59:49 brbri1 sudo[1283]: brbri : TTY=ttty ; PWD=/home/brbri1 ; USER=root ; COMMAND=/usr/bin/apt update
Mar 22 23:59:49 brbri1 sudo[1283]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 23 00:00:02 brbri1 sudo[1293]: pam_unix(sudo:session): session closed for user root
Mar 23 00:03:16 brbri1 systemd[1190]: launchpadlib-cache-clean.service - Clean up old files in the Launchpadlib cache was skipped because of an unset condition.
Mar 23 00:03:47 brbri1 sudo[1620]: brbri : TTY=ttty ; PWD=/home/brbri1 ; USER=root ; COMMAND=/usr/bin/apt upgrade
Mar 23 00:03:47 brbri1 sudo[1620]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 23 00:04:29 brbri1 sudo[1620]: pam_unix(sudo:session): session closed for user root
Mar 23 00:05:05 brbri1 sudo[2195]: brbri : TTY=ttty ; PWD=/home/brbri1 ; USER=root ; COMMAND=/usr/bin/apt install httpd -y
Mar 23 00:05:08 brbri1 sudo[2195]: pam_unix(sudo:session): session opened for user root(uid=0) by brbri1(uid=1000)
Mar 23 00:05:09 brbri1 sudo[2195]: pam_unix(sudo:session): session closed for user root

[9]+ Stopped                  journalctl -b
brbri1@brbri1:~$ journalctl -p err
Hint: You are currently not seeing messages from other users and the system.
      Users in groups 'adm', 'systemd-journal' can see all messages.
      Pass -q to turn off this notice.
-- No entries --
brbri1@brbri1:~$ journalctl -p err
Hint: You are currently not seeing messages from other users and the system.
      Users in groups 'adm', 'systemd-journal' can see all messages.
      Pass -q to turn off this notice.
-- No entries --
brbri1@brbri1:~$ journalctl -p err
Hint: You are currently not seeing messages from other users and the system.
      Users in groups 'adm', 'systemd-journal' can see all messages.
      Pass -q to turn off this notice.
-- No entries --
brbri1@brbri1:~$ sudo journalctl -p err
(sudo) password for brbri1:
Mar 22 23:56:58 localhost.localdomain kernel: ITS: WARNING: ITS mitigation depends on retpoline and rethunk support
Mar 22 23:56:43 localhost.localdomain kernel: vmx:rs:0000000000000000: [0x0] #ERR00R vmx:rs:0000000000000000: [0x0] #ERR00R This configuration is likely broken.
Mar 22 23:56:43 localhost.localdomain kernel: vmx:rs:0000000000000000: [0x0] #ERR00R Please switch to a supported graphics device to avoid problems.
Mar 22 23:57:45 brbri1 login[947]: PAM unable to dlopen(/usr/lib/security/pam_lastlog.so): cannot open shared object file: No such file or directory
Mar 22 23:57:45 brbri1 login[947]: PAM adding faulty module: pam_lastlog.so
lines 1-6/6 (END)
[10]+ Stopped                  sudo journalctl -p err
brbri1@brbri1:~$
```

➤ Sudo journalctl -u nginx

```
brbri1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

lines 1-6/6 (END)
[10]+ Stopped                  sudo journalctl -p err
brbri1@brbri1:~$ sudo journalctl -u nginx
-- No entries --
brbri1@brbri1:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 565 kB of archives.
After this operation, 1,596 kB of additional disk space will be used.
Get1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.6 [43.5 kB]
Get2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.6 [521 kB]
Fetched 565 kB in 2s (241 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 84831 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.6_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.6) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.6_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.6) ...
Setting up nginx-common (1.24.0-2ubuntu7.6) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.6) ...
* upgrading binary nginx
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
brbri1@brbri1:~$ sudo journalctl -u nginx
Mar 23 00:59:11 brbri1 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Mar 23 00:59:12 brbri1 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
brbri1@brbri1:~$
```

➤ Simulation d'incident :

```

brbl@brbl11:~$ sudo systemctl stop nginx
brbl@brbl11:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server.
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: inactive (dead) since Mon 2025-03-23 01:04:02 UTC; 2ss ago
   Duration: 44in 50,074s
   Docs: man:nginx(8)
  Process: 2571 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on: master_process on: (code=exited, status=0/SUCCESS)
  Process: 2573 ExecStart=/usr/sbin/nginx -g daemon on: master_process on: (code=exited, status=0/SUCCESS)
  Process: 2692 ExecStop=/usr/sbin/rtkit-stop-daemon --quiet --stop --retry QUIT/5 --pidfile /run/nginx.pid (code=exited, status=0/SUCCESS)
 Main PID: 2692 (code=exited, status=0/SUCCESS)
    CPU: 71ms

Mar 23 00:59:11 brbl11 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Mar 23 00:59:12 brbl11 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
Mar 23 01:04:02 brbl11 systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server...
Mar 23 01:04:02 brbl11 systemd[1]: nginx.service: Deactivated successfully.
Mar 23 01:04:02 brbl11 systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server.
brbl@brbl11:~$ sudo systemctl start nginx
brbl@brbl11:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server.
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-03-23 01:06:17 UTC; 10s ago
   Docs: man:nginx(8)
  Process: 2700 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on: master_process on: (code=exited, status=0/SUCCESS)
  Process: 2709 ExecStart=/usr/sbin/nginx -g daemon on: master_process on: (code=exited, status=0/SUCCESS)
 Main PID: 2711 (nginx)
    Tasks: 3 (limit: 4601)
   Memory: 2.4M (peak: 2.5M)
      CPU: 46ms
     CGroup: /system.slice/nginx.service
             └─2711 "nginx: master process /usr/sbin/nginx -g daemon on: master_process on:"
               └─2712 "nginx: worker process"
                 └─2713 "nginx: worker process"

Mar 23 01:06:36 brbl11 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Mar 23 01:06:37 brbl11 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
brbl@brbl11:~$

```

➤ Installation et configuration de Logrotate :

```

Process: 2709 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
Main PID: 2711 (nginx)
Tasks: 3 (limit: 4601)
Memory: 2.4M (peak: 2.5M)
CPU: 46ms
CGroup: /system.slice/nginx.service
└─2711 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
   └─2712 "nginx: worker process"
      └─2713 "nginx: worker process"

Mar 23 01:06:36 brrb11 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Mar 23 01:06:37 brrb11 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
brrb1@brrb11:~$ grep -s "/admin" /var/log/nginx/access.log
grep: /var/log/nginx/access.log: Permission denied
brrb1@brrb11:~$ sudo apt install logrotate -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
logrotate is already the newest version (3.21.0-2build1).
logrotate set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
brrb1@brrb11:~$ cat /etc/logrotate.conf
# see "man logrotate" for details

# global options do not affect preceding include directives

# rotate log files weekly

weekly

# use the adm group by default, since this is the owning group
# of /var/log/,
su root adm

# keep 4 weeks worth of backlogs
rotate 4

# create new (empty) log files after rotating old ones
create

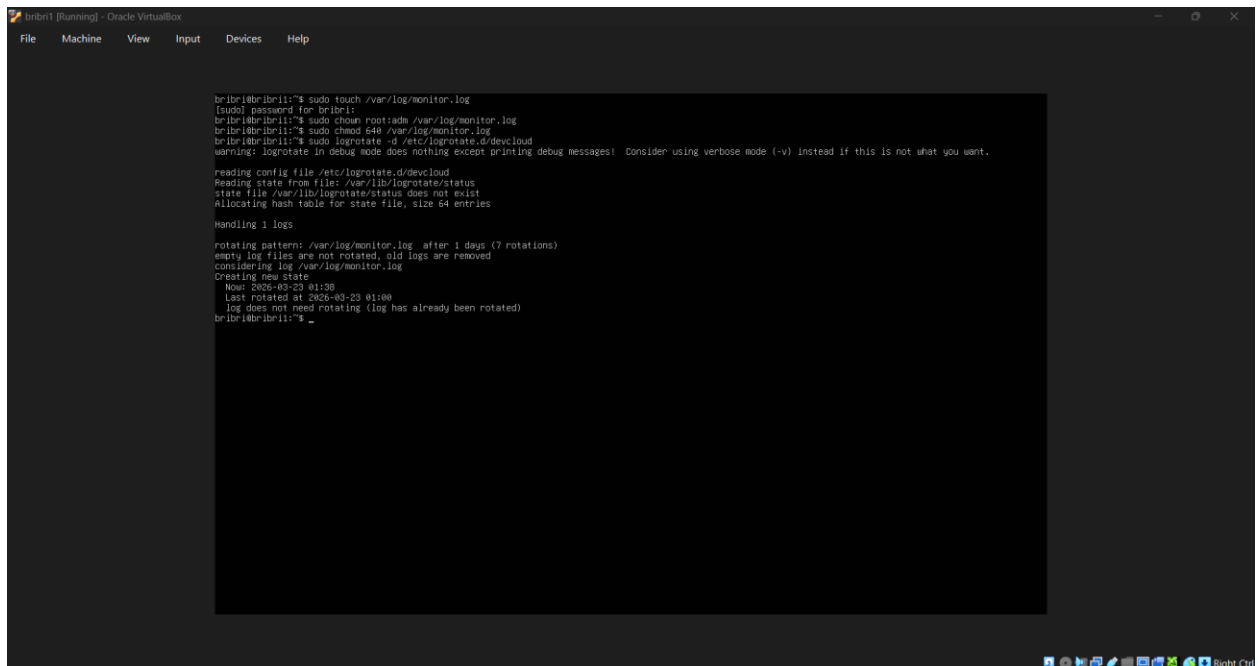
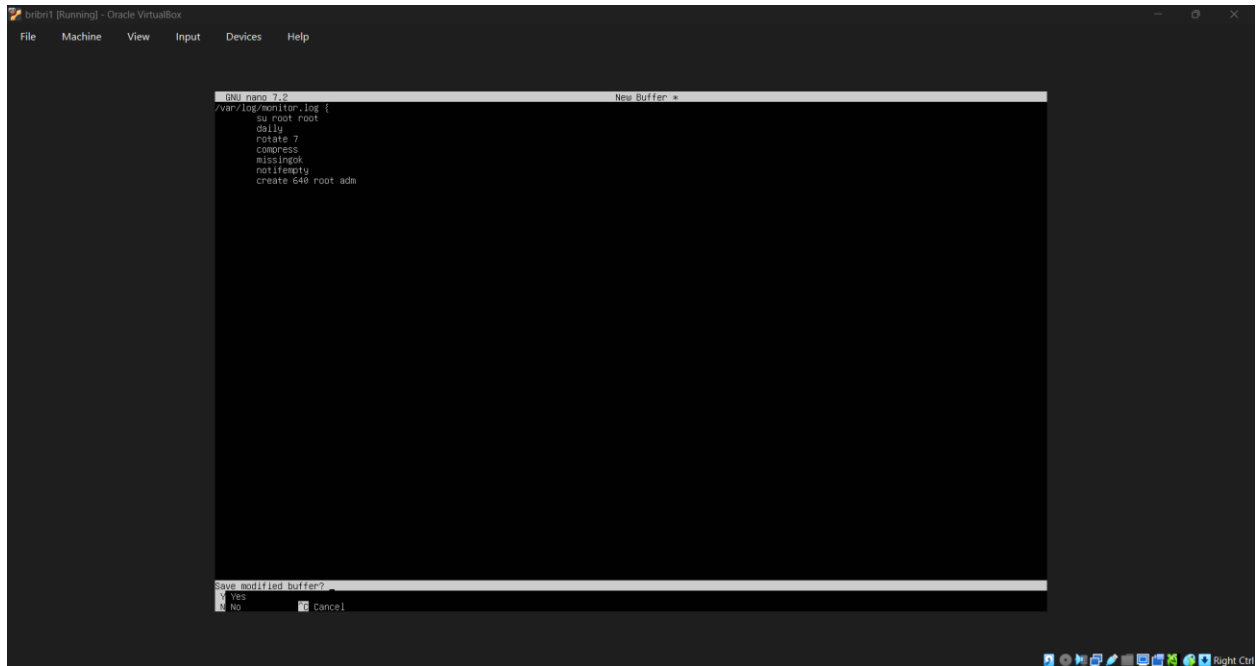
# use date as a suffix of the rotated file
dateext

# uncomment this if you want your log files compressed
#compress

# packages drop log rotation information into this directory
include /etc/logrotate.d

# system-specific logs may also be configured here.
brrb1@brrb11:~$ _

```

```
brbl1@brbl1:~$ sudo logrotate -v /etc/logrotate.d/devcloud
reading config file /etc/logrotate.d/devcloud
Creating state file /var/lib/logrotate/status
acquired lock on state file /var/lib/logrotate/status
Reading state from file: /var/lib/logrotate/status
Allocating hash table for state file, size 64 entries

Handling 1 logs

rotating pattern: /var/log/monitor.log after 1 days (7 rotations)
empty log files are not rotated, old logs are removed
considering log /var/log/monitor.log
Creating new state
Now: 2026-03-23 01:40
Last rotated at 2026-03-23 01:00
log does not need rotating (log has already been rotated)
brbl1@brbl1:~$ _
```

➤ Installation d'un outil de supervision :

```
Docs: https://httpd.apache.org/docs/2.4/
Process: 14692 ExecStart=/usr/sbin/apachectl start (code=exited, status=1/FAILURE)
CPU: 51ms

Mar 23 01:58:16 brbl1 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 23 01:58:16 brbl1 apachectl[14694]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using fd17:625c:f037:2:a003:2::1
ff4ed748de. Set the 'ServerName' directive globally to suppress this message
Mar 23 01:58:16 brbl1 apachectl[14694]: (98)Address already in use: AH00072: make_sock: could not bind to address [::]:80
Mar 23 01:58:16 brbl1 apachectl[14694]: (98)Address already in use: AH00072: make_sock: could not bind to address 0.0.0.0:80
Mar 23 01:58:16 brbl1 apachectl[14694]: no listening sockets available, shutting down
Mar 23 01:58:16 brbl1 apachectl[14694]: AH00015: Unable to open logs
Mar 23 01:58:16 brbl1 systemd[1]: apache2.service: Control process exited, code=exited, status=1/FAILURE
Mar 23 01:58:16 brbl1 systemd[1]: apache2.service: Failed with result 'exit-code'.
Mar 23 01:58:16 brbl1 systemd[1]: Failed to start apache2.service - The Apache HTTP Server.
Scanning processes...
Scanning Linux slaves...

Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
brbl1@brbl1:~$ sudo systemctl stop nginx
brbl1@brbl1:~$ sudo systemctl restart apache2
brbl1@brbl1:~$ sudo systemctl status apache2
* apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-03-23 01:59:14 UTC; 29s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 14793 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 14796 (apache2)
       Tasks: 6 (limit: 4901)
      Memory: 12.3M (peak: 12.6M)
         CPU: 124ms
    CGroup: /system.slice/apache2.service
            └─14796 /usr/sbin/apache2 -k start
              └─14798 /usr/sbin/apache2 -k start
                └─14799 /usr/sbin/apache2 -k start
                  └─14800 /usr/sbin/apache2 -k start
                    └─14801 /usr/sbin/apache2 -k start
                      └─14802 /usr/sbin/apache2 -k start

Mar 23 01:59:13 brbl1 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 23 01:59:14 brbl1 apachectl[14795]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using fd17:625c:f037:2:a003:2::1
Mar 23 01:59:14 brbl1 systemd[1]: Started apache2.service - The Apache HTTP Server.
https://302020.000
```

```
brbr11 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Active: active (running) since Mon 2020-03-23 01:59:14 UTC; 29s ago
Docs: https://httpd.apache.org/docs/2.4/
Process: 14793 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
Main PID: 14796 (apache2)
Tasks: 6 (limit: 4601)
Memory: 12.5M (peak: 12.0M)
CPU: 124ms
CGroup: /system.slice/apache2.service
└─14796 /usr/sbin/apache2 -k start
   14798 /usr/sbin/apache2 -k start
   14799 /usr/sbin/apache2 -k start
   14800 /usr/sbin/apache2 -k start
   14801 /usr/sbin/apache2 -k start
   14802 /usr/sbin/apache2 -k start

Mar 23 01:59:13 brbr11 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 23 01:59:14 brbr11 apachectl[14795]: AH00558: apache2: could not reliably determine the server's fully qualified domain name, using fd17:625c:f037:2::a00::1
Mar 23 01:59:14 brbr11 systemd[1]: Started apache2.service - The Apache HTTP Server.
[11]# Stopped
brbr11@brbr11:~$ sudo systemctl status apache2
brbr11@brbr11:~$ sudo systemctl status nagios4
* nagios4.service - nagios4
Loaded: loaded (/usr/lib/systemd/system/nagios4.service; disabled; preset: enabled)
Active: active (running) since Mon 2020-03-23 01:59:01 UTC; 10min ago
Docs: man:nagios4
Main PID: 13891 (nagios4)
Tasks: 6 (limit: 4601)
Memory: 2.5M (peak: 4.9M)
CPU: 65ms
CGroup: /system.slice/nagios4.service
└─13891 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg
   13893 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
   13894 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
   13895 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
   13896 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
   13897 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg

Mar 23 01:59:01 brbr11 nagios4[13891]: Successfully launched command file worker with pid 13897
Mar 23 01:57:00 brbr11 nagios4[13891]: SERVICE ALERT: localhost:Swap Usage:CRITICAL:SOFT:1:SWAP CRITICAL - 0% free (0MB out of 0MB) - Swap is either disabled.
Mar 23 01:57:00 brbr11 nagios4[13891]: SERVICE ALERT: localhost:SSH:CRITICAL:SOFT:1:connect to address 127.0.0.1 and port 22: Connection refused
Mar 23 01:56:00 brbr11 nagios4[13891]: SERVICE ALERT: localhost:Swap Usage:CRITICAL:SOFT:2:SWAP CRITICAL - 0% free (0MB out of 0MB) - Swap is either disabled.
Mar 23 01:56:00 brbr11 nagios4[13891]: SERVICE ALERT: localhost:SSH:CRITICAL:SOFT:2:connect to address 127.0.0.1 and port 22: Connection refused
Mar 23 01:53:00 brbr11 nagios4[13891]: SERVICE ALERT: localhost:SSH:CRITICAL:SOFT:3:connect to address 127.0.0.1 and port 22: Connection refused
Mar 23 01:53:00 brbr11 nagios4[13891]: SERVICE ALERT: localhost:SSH:CRITICAL:SOFT:3:connect to address 127.0.0.1 and port 22: Connection refused
Mar 23 02:00:00 brbr11 nagios4[13891]: SERVICE ALERT: localhost:SSH:CRITICAL:HARD:4:connect to address 127.0.0.1 and port 22: Connection refused
Mar 23 02:00:00 brbr11 nagios4[13891]: SERVICE NOTIFICATION: nagiosadmin:localhost:Swap Usage:CRITICAL:notify-service-by-email:SWAP CRITICAL - 0% free (0MB out of 0MB)
Mar 23 02:00:00 brbr11 nagios4[13891]: SERVICE ALERT: localhost:Swap Usage:CRITICAL:HARD:4:SWAP CRITICAL - 0% free (0MB out of 0MB) - Swap is either disabled.
[12]# Stopped
brbr11@brbr11:~$ sudo systemctl status nagios4
brbr11@brbr11:~$
```



```

Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  fonts-dejavu fonts-dejavu-extra fping libevent-2.1-7t64 libevent-pthreads-2.1-7t64 libltdl7 libmodbus5 libodbc2
  libopenipmi0t64 mysql-client mysql-client-8.0 mysql-client-core-8.0 php-bcmath php-gd php-ldap php-mbstring
  php-mysql php-xml php8.3-bcmath php8.3-gd php8.3-ldap php8.3-mbstring php8.3-mysql php8.3-xml snmpd
Paquets suggérés :
  odbc-postgresql tdsodbc snmptrapd zabbix-nginx-conf virtual-mysql-server
Les NOUVEAUX paquets suivants seront installés :
  fonts-dejavu fonts-dejavu-extra fping libevent-2.1-7t64 libevent-pthreads-2.1-7t64 libltdl7 libmodbus5 libodbc2
  libopenipmi0t64 mysql-client mysql-client-8.0 mysql-client-core-8.0 php-bcmath php-gd php-ldap php-mbstring
  php-mysql php-xml php8.3-bcmath php8.3-gd php8.3-ldap php8.3-mbstring php8.3-mysql php8.3-xml snmpd zabbix-agent
  zabbix-apache-conf zabbix-frontend-php zabbix-server-mysql
0 mis à jour, 29 nouvellement installés, 0 à enlever et 21 non mis à jour.
Il est nécessaire de prendre 15,6 Mo dans les archives.
Après cette opération, 122 Mo d'espace disque supplémentaires seront utilisés.
Réception de :1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 snmpd amd64 5.9.4+dfsg-1.1ubuntu3.2 [59,5 kB]
Réception de :2 http://archive.ubuntu.com/ubuntu noble/main amd64 libevent-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [145 kB]

```



QUESTIONS ?

1. Quelle est la différence entre les logs système et les logs applicatifs ?

La distinction réside principalement dans l'origine de l'événement enregistré :

Logs système : Ils sont produits directement par le noyau Linux et les services essentiels de l'OS. Ils servent à documenter la vie de la machine, comme l'activité du matériel (disque défaillant), les phases de démarrage ou les accès distants via SSH.

Logs applicatifs : Ils proviennent des logiciels tiers installés sur le serveur. Ils détaillent le fonctionnement interne de ces outils, par exemple les requêtes HTTP traitées par un serveur Nginx, l'exécution de requêtes SQL ou les alertes de supervision.

2. Pourquoi est-il nécessaire de centraliser les logs en entreprise ?

Le regroupement des journaux sur un serveur unique offre plusieurs bénéfices stratégiques :

Efficacité opérationnelle : Cela simplifie grandement le diagnostic en évitant de se connecter individuellement sur chaque machine ; tout est accessible au même endroit.

Sécurité accrue : La centralisation permet d'identifier des schémas d'attaque complexes en recoupant les données de plusieurs sources. De plus, cela protège les preuves : si un intrus efface les logs locaux pour masquer ses traces, les copies envoyées en temps réel au serveur central restent intactes.

Conformité légale : Il est plus facile de respecter les durées de conservation imposées par des normes comme le RGPD ou PCI-DSS via un archivage centralisé.

3. Quelles sont les conséquences d'un volume de logs trop important ?

Une croissance incontrôlée des fichiers logs peut mener à une saturation du stockage, entraînant :

L'impossibilité pour le système d'enregistrer de nouveaux événements.

Des dysfonctionnements applicatifs, certains logiciels refusant de s'exécuter si l'espace disque est nul.

Une instabilité générale du système d'exploitation, pouvant aller jusqu'au plantage complet.

4. Quel est le rôle de l'outil Logrotate ?

Logrotate a pour mission d'automatiser la maintenance des fichiers de logs afin d'économiser l'espace disque. Son action se décline en quatre étapes :

La rotation : Il archive le fichier actuel et en commence un nouveau.

La compression : Il réduit la taille des archives (souvent au format .gz).

La suppression : Il élimine les fichiers les plus anciens selon un quota défini.

La planification : Il exécute ces tâches périodiquement sans intervention humaine.

5. Pourquoi la surveillance de la RAM et du CPU est-elle primordiale ?

Le monitoring de ces ressources est le garant de la santé de l'infrastructure:

Performance et stabilité : Une saturation CPU ralentit les traitements, tandis qu'un manque de RAM force le système à utiliser le disque (swap), ce qui fait chuter les performances.

Détection d'anomalies : Surveiller ces métriques aide à repérer des fuites de mémoire ou des processus malveillants, comme le minage de crypto monnaies non autorisé.

Anticipation : Ces données permettent de planifier les besoins futurs en matériel et d'assurer une disponibilité constante des services pour les utilisateurs.

Conclusion

Ce TD m'a permis de maîtriser la surveillance en temps réel avec htop et l'analyse précise des journaux via journalctl. J'ai compris l'importance d'automatiser la gestion de l'espace disque avec Logrotate et de centraliser la supervision avec des outils comme Nagios ou Zabbix. Ces compétences sont, pour moi, les piliers d'une administration système stable, sécurisée et réactive face aux incidents.